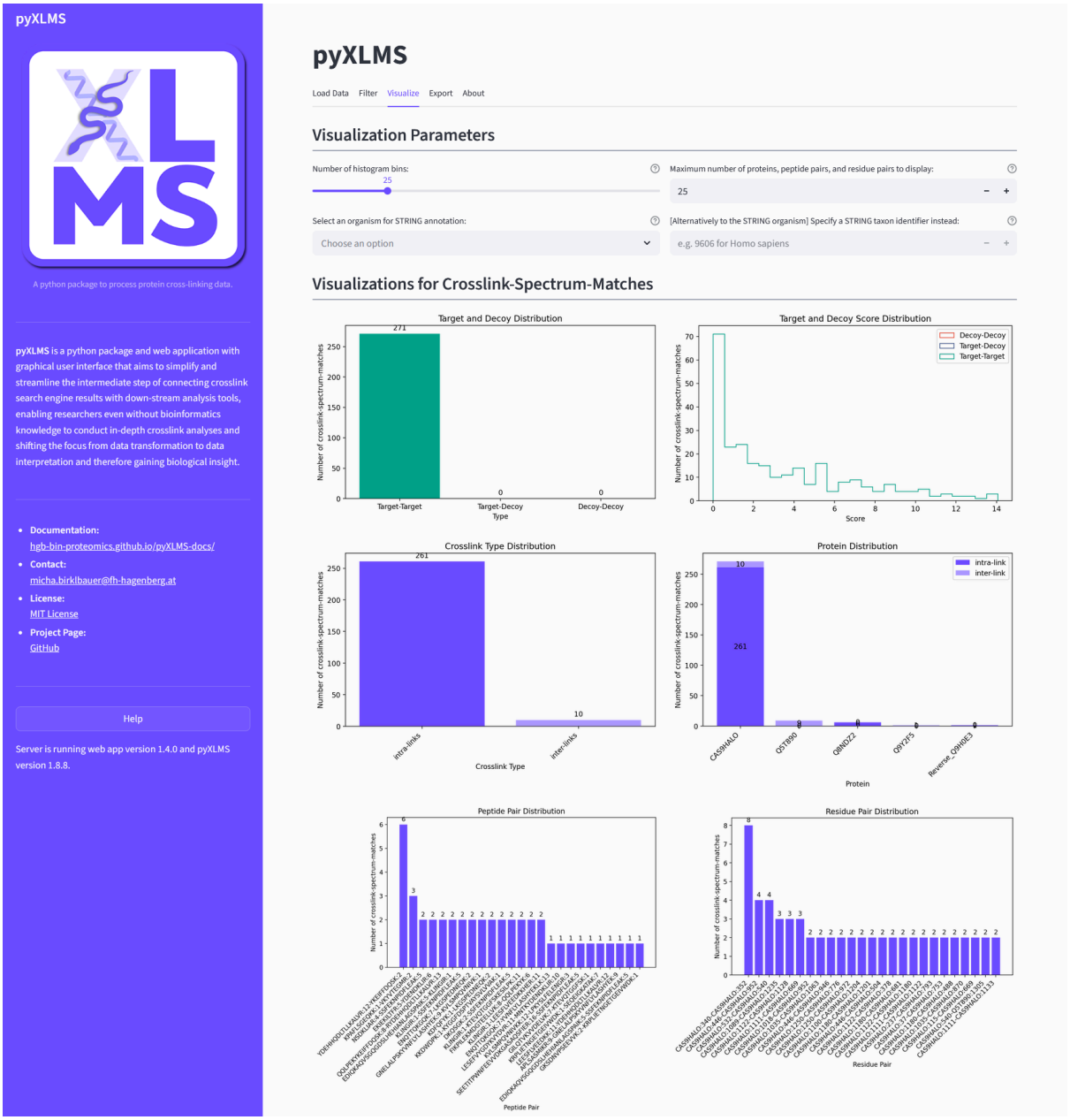
Supplementary Figure 1



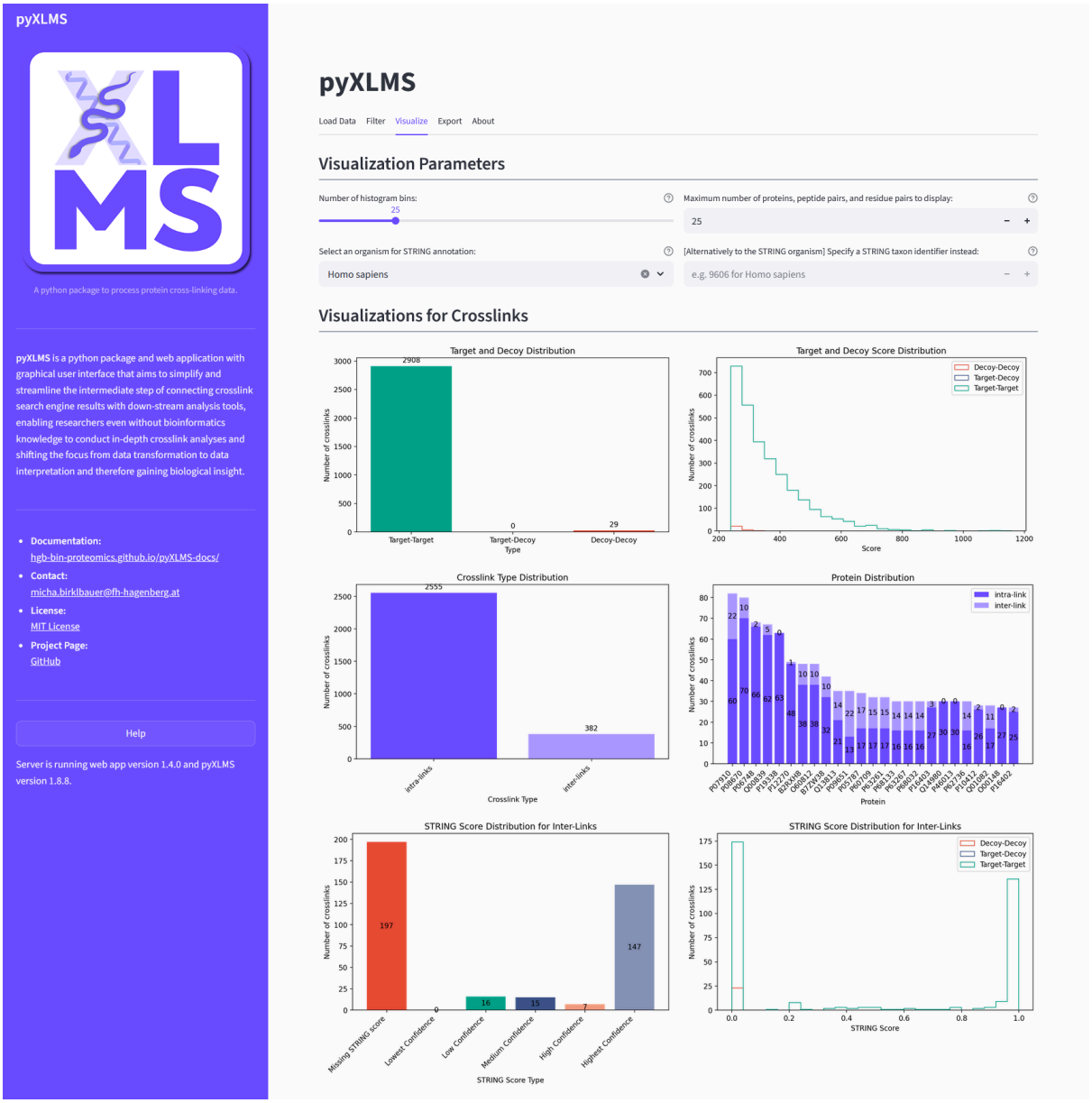
Supplementary Figure 1: *Overview of the visualization tab in the pyXLMS web application for crosslink-spectrum-matches (CSMs). Users can control the visualization, e.g. the number of histogram bins and the number of shown proteins/peptide pairs/residue pairs via the parameters at the top. The first panel shows an overview of the target-target, target-decoy, and decoy-decoy matches which in this case only shows a bar for target-target CSMs because the results were already validated and decoy free. The second panel shows the target-decoy score distribution. The third panel shows the number of intra- and inter-link CSMs, while*

the fourth panel shows the number of intra- and inter-link CSMs per protein. The fifth panel shows the number of CSMs per unique peptide pair, while the number of CSMs per unique residue pair is shown in panel six. Visualizations are shown for validated results of Scout using replicate one of the Cas9 dataset.

This figure can be reproduced by uploading the file (Crosslink Search Engine: Scout, Crosslinker: DSBSO) to the web app:

https://github.com/hgb-bin-proteomics/pyXLMS-manuscript/blob/master/results/scout/Cas9_HeLa_Cyt_r1/Cas9_HeLa_Cyt_r1_filtered_CSMs.csv

Supplementary Figure 2



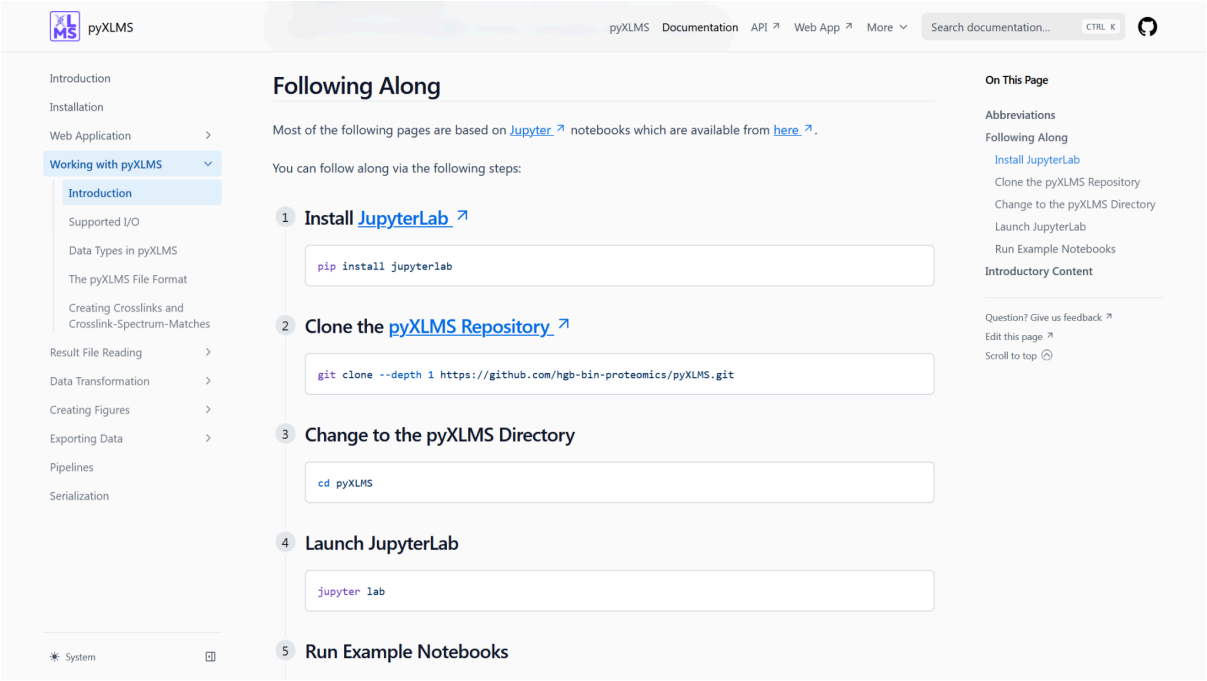
Supplementary Figure 2: *Overview of the visualization tab in the pyXLMS web application for crosslinks (XLs, in this case unique residue pairs). The first panel shows an overview of the target-target, target-decoy, and decoy-decoy XLs. The second panel shows the target-decoy score distribution. The third panel shows the number of intra- and inter-link XLs, while the fourth panel shows the number intra- and inter-link XLs per protein. Panel five shows the STRING [1] (main) score distribution for all interprotein XLs, sorted into the confidence classes suggested by STRING. In the sixth panel the STRING score is displayed as a distribution for target-target, target-decoy, and decoy-decoy XLs showing that decoy XLs all are*

assigned a zero STRING score which indicates that the protein-protein interaction does not exist in STRING. Visualizations are shown for results validated for 1% estimated residue pair-level FDR using data of human K562 nuclei from the same study as used in the main manuscript [2].

This figure can be reproduced by uploading the file (Crosslink Search Engine: Custom, Crosslinker: DSBSO) to the web app (STRING organism “Homo sapiens” has to be selected in the Visualization tab):

https://github.com/hgb-bin-proteomics/pyXLMS-manuscript/blob/master/results/string/Nucleus_Rep1_Crosslinks.parquet

Supplementary Figure 3



Supplementary Figure 3: *An excerpt of the pyXLMS user guide available via https://hgb-bin-proteomics.github.io/pyXLMS-docs. The pyXLMS user guide features step-by-step instructions for using the pyXLMS python package as well as the pyXLMS web application, including many common usage examples and exemplary data to try all the functionality first hand.*

Supplementary Section 1 - pyXLMS Quick Links

Here is a quick overview of the most important pyXLMS resources:

- Web Application: https://hgb-bin-proteomics.github.io/pyXLMS-app
- User Guide: https://hgb-bin-proteomics.github.io/pyXLMS-docs
- GitHub Repository: https://github.com/hgb-bin-proteomics/pyXLMS

Here is a complete overview of all available pyXLMS resources:

- https://github.com/hgb-bin-proteomics/pyXLMS: pyXLMS git repository, source code, and main page
- https://hgb-bin-proteomics.github.io/pyXLMS-app: publicly hosted instance of the pyXLMS web application that is free to use for everyone
- https://hgb-bin-proteomics.github.io/pyXLMS-docs: User guide with in-depth step-by-step instructions for using both the python package and the web application
- https://hgb-bin-proteomics.github.io/pyXLMS/: pyXLMS python package documentation
- https://github.com/hgb-bin-proteomics/pyXLMS/tree/master/gui: source code of the pyXLMS web application
- https://github.com/hgb-bin-proteomics/pyXLMS-app: information about our publicly hosted instance of the pyXLMS web application
- https://github.com/hgb-bin-proteomics/pyXLMS-template: a template repository for projects using pyXLMS
- https://github.com/hgb-bin-proteomics/pyXLMS-docs: source code of the user guide website
- https://github.com/hgb-bin-proteomics/pyXLMS-manuscript: source code for all analyses of the manuscript

Supplementary Section 2 - pyXLMS Web App Feature Support

The web app supports almost all functionality of the pyXLMS python package. However, the following features are **exclusively available via the python package**:

- Some parser parameters are only available in the python package.
- Processing of search results from more than one search engine and crosslinker at the same time.
- Creating intersections of search results from more than one search engine or crosslinker. You can read more about how to do that with the python package here. We also used that for the pyXLMS manuscript to create intersections of different replicates and search engines, you can find the code for that here.
- Creating Venn diagrams. You can read more about how to do that with the python package here. We also used that for the pyXLMS manuscript, you can find the code for that here.
- Annotating estimated false-discovery-rate (FDR), more about that here.
- Re-Annotating decoy labels is only available in the python package, see more here.
- Some filters are only available in the python package, e.g. filtering by peptide or residue pair. You can read more here.
- Export to ProForma is only available in the python package as described here.

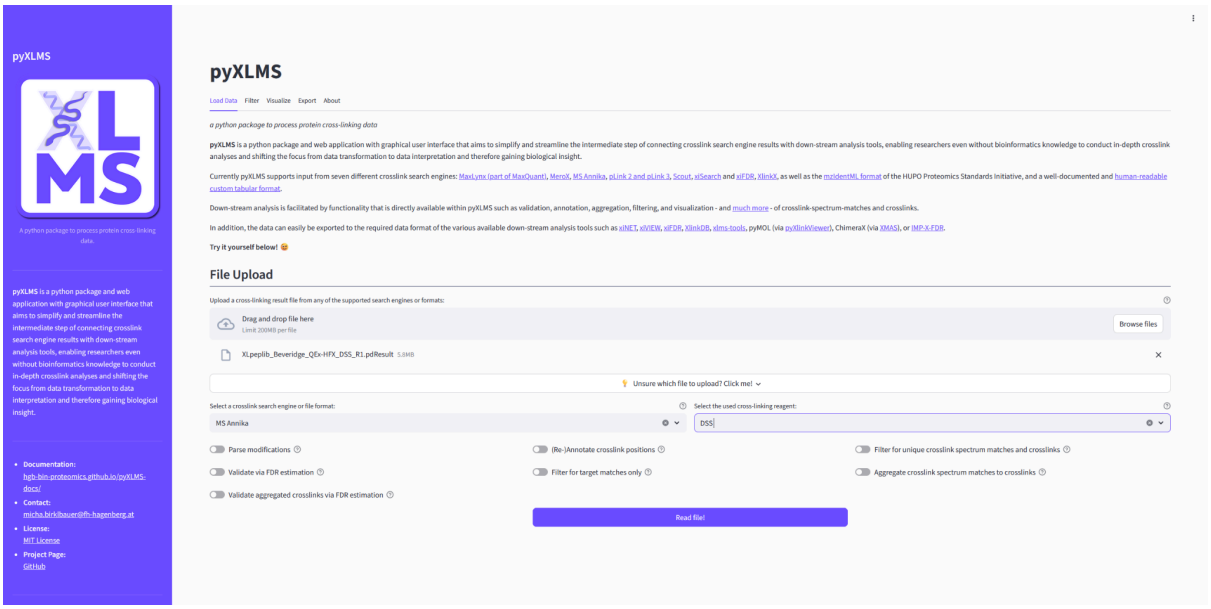
TIP

The file size limit is 200 MB in the web app. For self-hosted instances this can be increased via the config using the `maxUploadSize` option. For more info see here. However, we generally recommend to run anything with more than 50 000 crosslink-spectrum-matches/crosslinks via the python package. The python package shows progress bars for every operation and is therefore favorable for longer operations like validation or re-annotation.

Supplementary Section 3 - pyXLMS Web App User Manual

This guide can also be found at https://pyxlms.dev/docs/webapp.

Loading Data



File Selection

The first step when using the pyXLMS web application is that you need to upload one or more result files containing crosslink-spectrum-matches (CSMs) or crosslinks/residue pairs (XLs) in the Load Data tab. You can find a list of supported crosslink search engines and input formats here.

We also provide some example files via the pyXLMS GitHub repository that you can try!

TIP

Starting from pyXLMS v1.6.1 the web application also supports input of multiple result files, e.g. if you have CSMs and XLs from the same run, result files of different fractions of your sample, or replicates.

Selecting Crosslink Search Engine or File Format

Next you need to select the correct crosslink search engine or file format from the dropdown menu. This should simply be the crosslink search engine that you used for crosslink identification with your mass spectrometry data, or a generic format like mzIdentML.

Selecting Crosslink Reagent

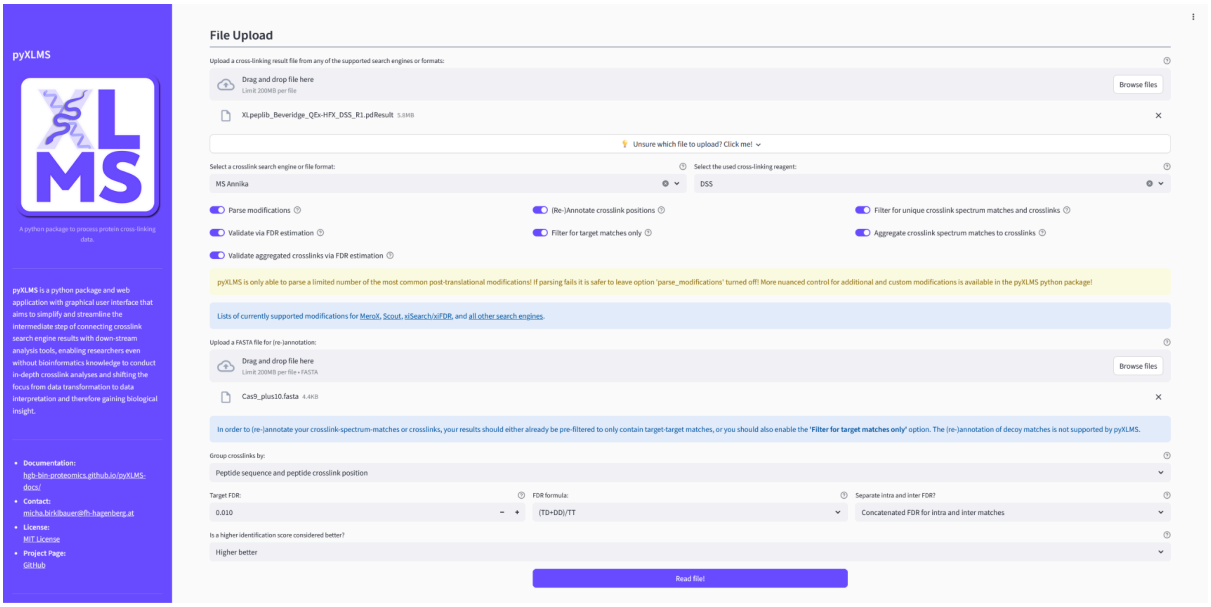
Then you need to select the crosslinking reagent that was used in your experiment from the next dropdown menu. If your reagent is not in the list you can select Custom and two additional input fields will pop up where you can specify a crosslinker name and the monoisotopic delta mass of the crosslink modification.

Reading the Results

If you don't want to employ any of the additional and optional pre-processing steps (see next section) you can read your results now by clicking on the Read file(s)! button!

[Optional] Pre-Processing Results

pyXLMS offers several different pre-processing steps as outlined below.



[Optionally] Apply One or More Pre-Processing Steps

- **Parse Modifications**

  Enabling this option will try to parse post-translation-modifications (PTMs) of your CSMs from your result files.

  NOTE

  Please note that the pyXLMS web app only supports a limited set of the most common PTMs! If parsing fails it is better to leave this option turned off and use the pyXLMS python package directly which gives a more nuanced control over modifications!

- **Re-Annotate Crosslink Positions**

  This option will allow you to upload a FASTA file to re-annotate the protein crosslink positions of your CSMs and XLs based on the given FASTA file. This might be useful if the annotation is missing because it was not given by the crosslink search engine, or if the crosslink search engine annotated something wrongly.

- **Filtering for Unique CSMs and XLs**

  If turned on this option will filter out any non-unique CSMs and XLs from your results. A crosslink is considered unique if its sequence and peptide crosslink positions are unique (option 1) or if its protein crosslink positions are unique (option 2) - these options are controlled via the Group crosslinks by selector.
- **Validation via FDR Estimation**

  Enabling this option will validate your results for a given target false-discover-rate (FDR) via FDR estimation and will only keep CSMs and XLs that fall within the target FDR.

  IMPORTANT

  We recommend uploading results that are already validated in the crosslink search engine of your choice. Validation in pyXLMS uses a generic target-decoy approach with a single score that might not yield the same results as the search engine specific validation!
- **Filtering for Target Matches**

  If applied this option will filter out any CSMs and XLs that are not target-target matches.
- **Aggregation of CSMs to XLs**

  Whether or not your CSMs (if there are any) should be aggregated to XLs.
- **Validation of Aggregated XLs**

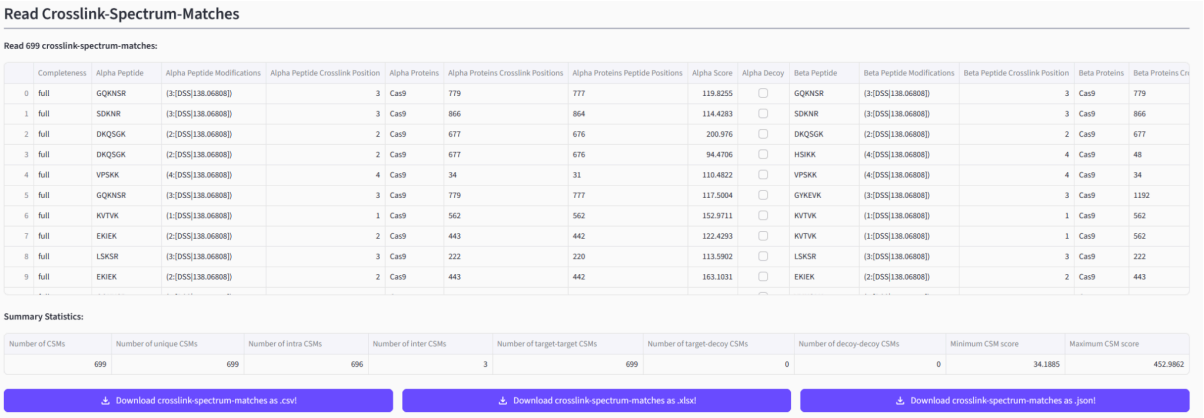
  Controls if your aggregated XLs from the previous pre-processing step should also be validated via FDR estimation. Shares the same settings as Validation via FDR Estimation.

Reading the Results

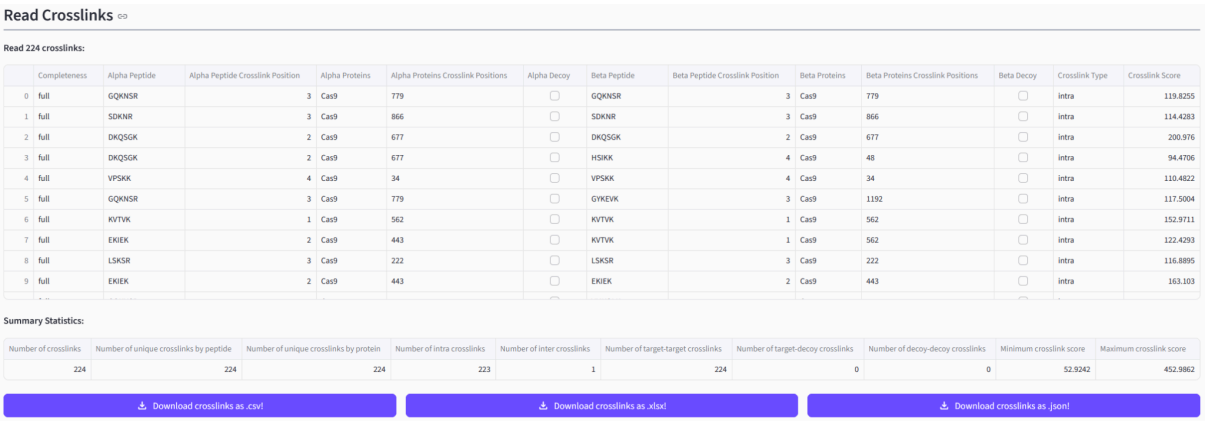
You can read your results now by clicking on the Read file(s)! button!

Inspecting Your Results

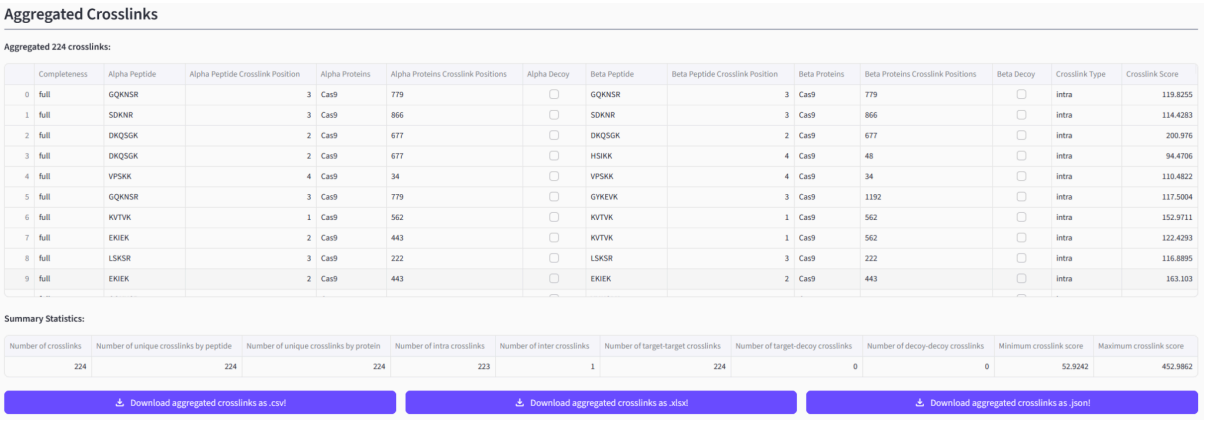
Depending on the result file(s) you uploaded and the selected pre-processing steps you will see the read CSMs and/or XLs and/or aggregated XLs. The web app will both display the read data itself as well as a short summary statistic. You will also be able to directly download the data in different formats.



➡ Exemplary depiction of read CSMs.



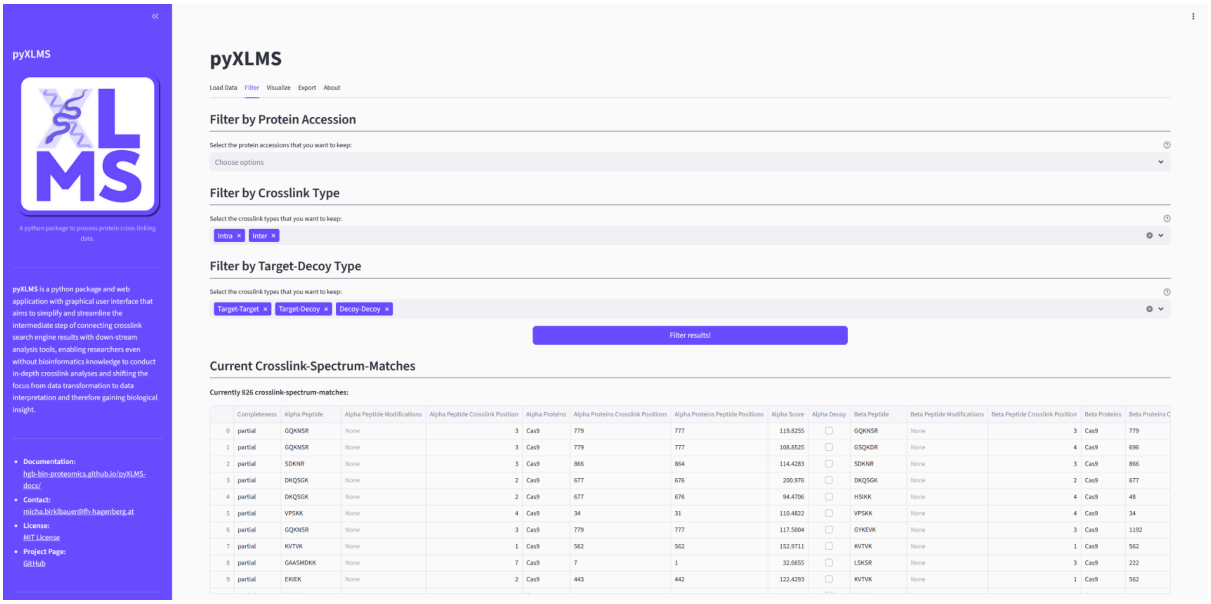
➡ Exemplary depiction of read XLs.



➡ Exemplary depiction of aggregated XLs created from the read CSMs.

Filtering Results

The Filter tab offers some additional filtering options:



Filter Options

Filter by Protein Accession

Select your proteins of interest from the dropdown menu. Any CSM or XL that does not have at least one peptide corresponding to one of the selected proteins will be filtered out. Leaving this option blank will not apply the filter.

Filter by Crosslink Type

Select which type of crosslinks you are interested in. Leaving this option blank will not apply the filter.

Filter by Target-Decoy Type

Select which database match types you are interested in. Option Target-Decoy corresponds to both "Target-Decoy" and "Decoy-Target" matches. Leaving this option blank will not apply the filter.

IMPORTANT

Please note that this filter will remove any CSMs and XLs that have missing target-decoy labels!

Inspect Your Filtered Results

After hitting the Filter results! button the data at the bottom of the page will update and display your current CSMs and/or XLs and/or aggregated XLs after the filtering.

Visualizing Results

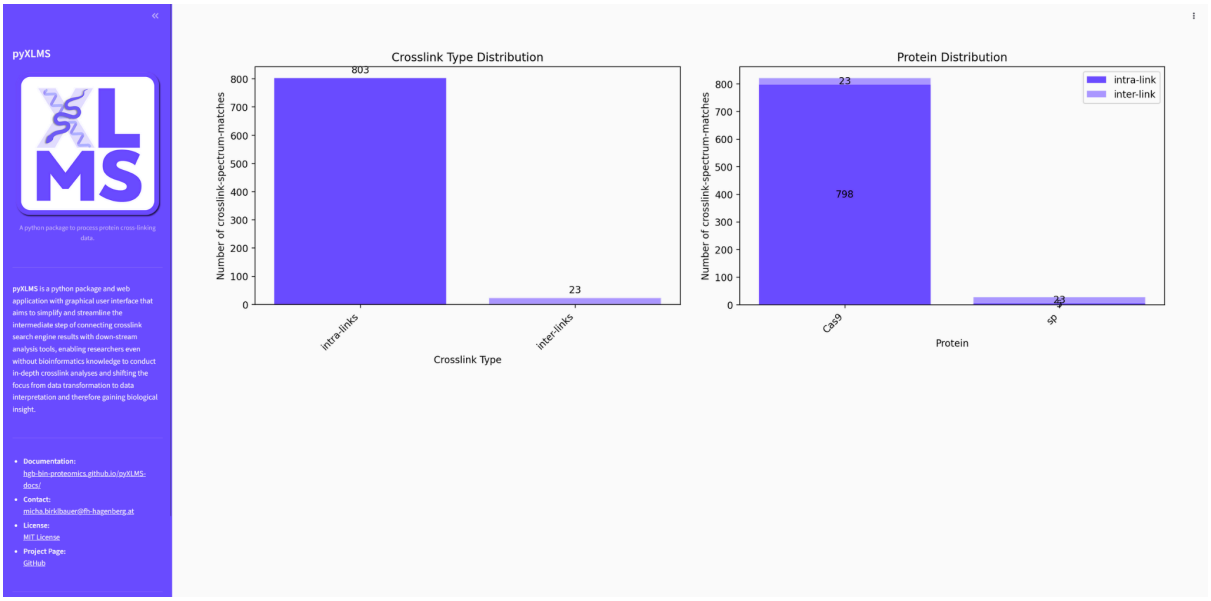
The Visualize tab will show some basic plots for your CSMs and/or XLs and/or aggregated XLs.

NOTE

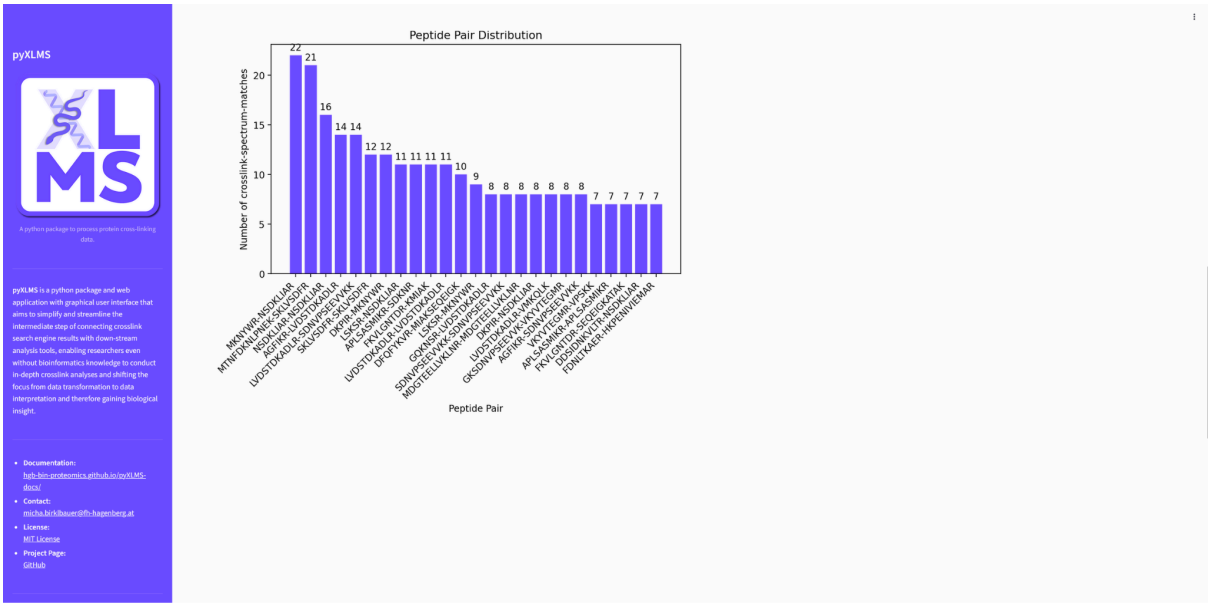
Please note that the visualizations are heavily influenced by your pre-processing and filtering steps! E.g. filtering for target matches will obviously mean that Target and Decoy plots will only show target-target matches!



- The Target and Decoy Distribution plot will show the number of "Target-Target", "Target-Decoy" and "Decoy-Decoy" matches in your results.
- The Target and Decoy Score Distribution plot will show the score distribution of "Target-Target", "Target-Decoy" and "Decoy-Decoy" matches in your results.



- The Crosslink Type Distribution plot will show the number of intra- and inter-links in your results.
- The Protein Distribution plot will show the top *n* proteins with the most crosslink evidence in terms of number of CSMs/XLs. It additionally shows the number of intra- and inter-links per protein. The *n* parameter is controlled via the Visualization Parameters section at the top of the page.



- The Peptide Pair Distribution plot is only available for CSMs and shows the most frequently observed peptide pairs in your CSMs.

Exporting Results

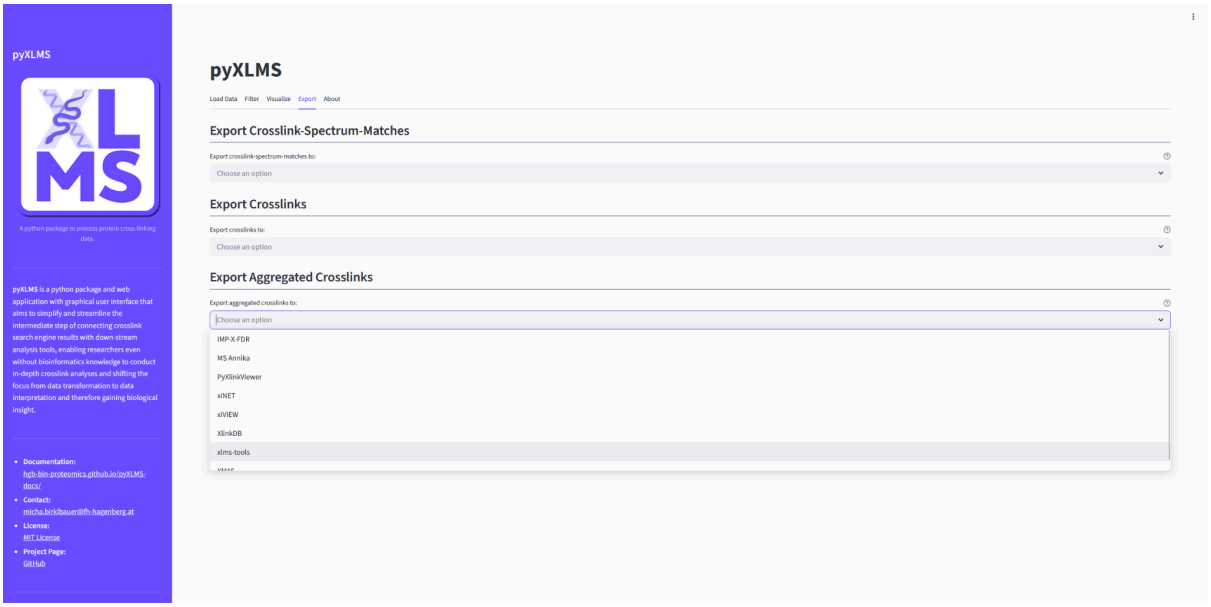
The Export tab allows you to export your current selection of CSMs and/or XLs and/or aggregated XLs to one of the supported down-stream analysis tools or export formats.

IMPORTANT

Please note that all exporters check if the required information to create a successful export is available at runtime! If key information is missing, an error will be raised!

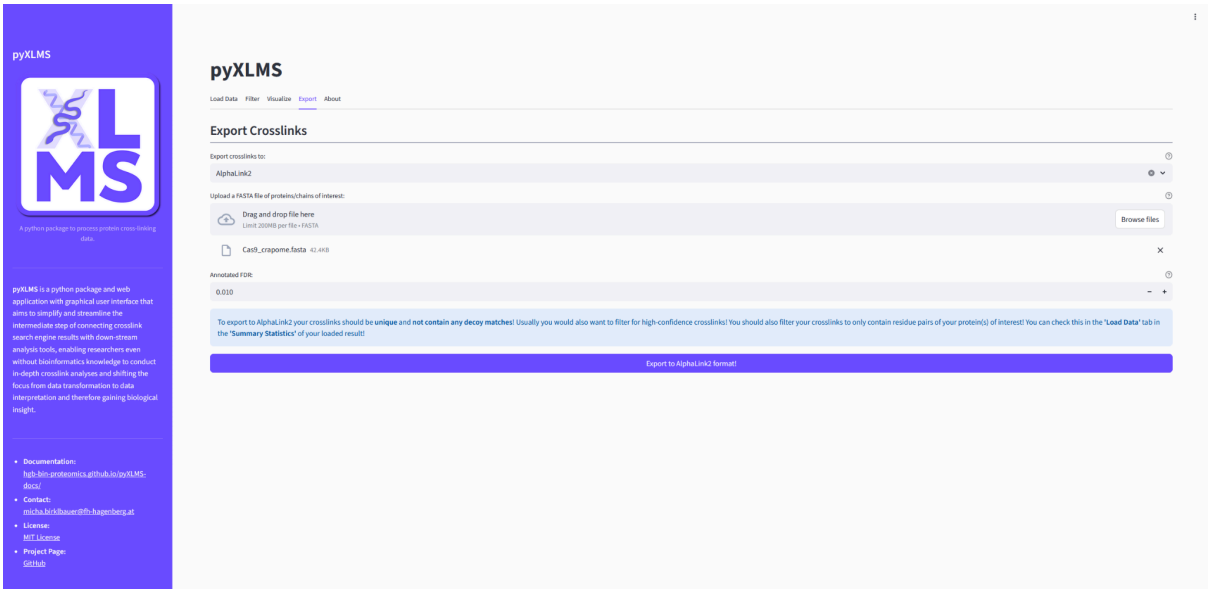
Exporting Your Results

Select the Down-Stream Analysis Tool/Export Format



Select the down-stream analysis tool or export format that you want to get.

Verify Your Pre-Processing and Filtering



After selecting an export option a blue callout will appear with recommendations on how your results should be pre-processed and filtered for the export to work correctly. This is not only important for the export but subsequent down-stream analysis, e.g. down-stream analysis tools might not work correctly if your data is not sufficiently pre-processed and filtered, even if the export worked flawlessly.

WARNING

Please note that the exporters do not check which pre-processing steps have been carried out beforehand! Some down-stream analysis tools might require validated results, some might require raw results, some might require target matches only, some might require target and decoy matches, etc. Which filtering steps are applied before export are up to you - the user - and your own responsibility. Please read the instructions of the down-stream analysis tool you want to use and get familiar with it and the required data before exporting! The exporter tutorial pages also give a short overview of recommended pre-processing steps!

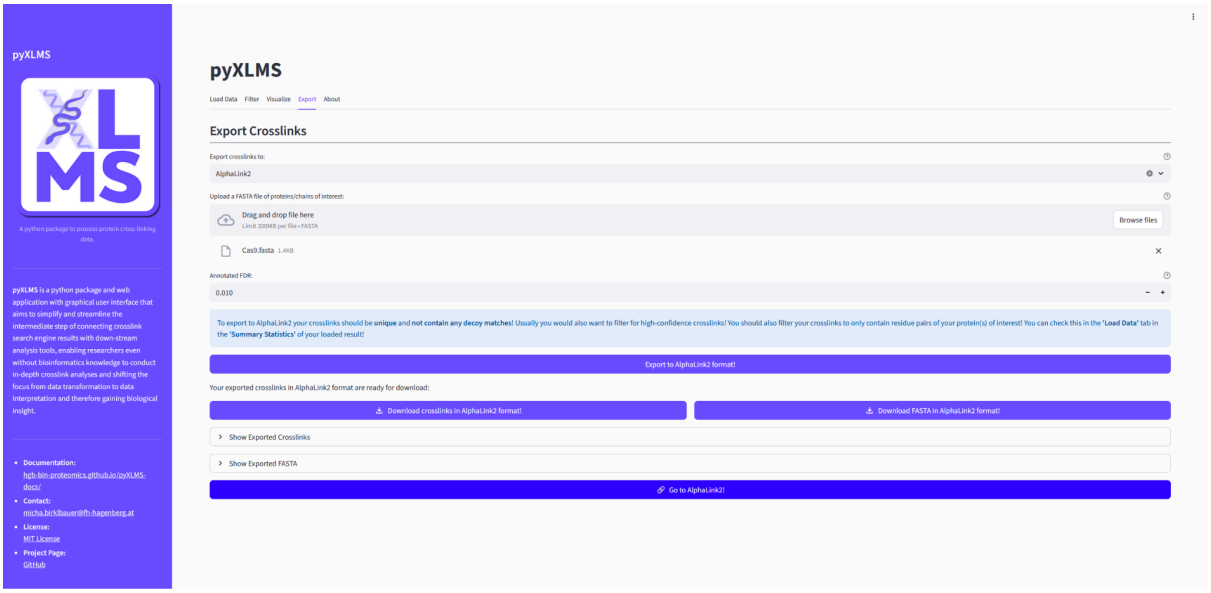
Supply Any Additionally Required Data

Some exporters need additional data to work, for example a FASTA file or other not yet known meta information. Please fill in this information.

Create the Export

Click the Export to * button to run the exporter.

Download the Exported File(s)



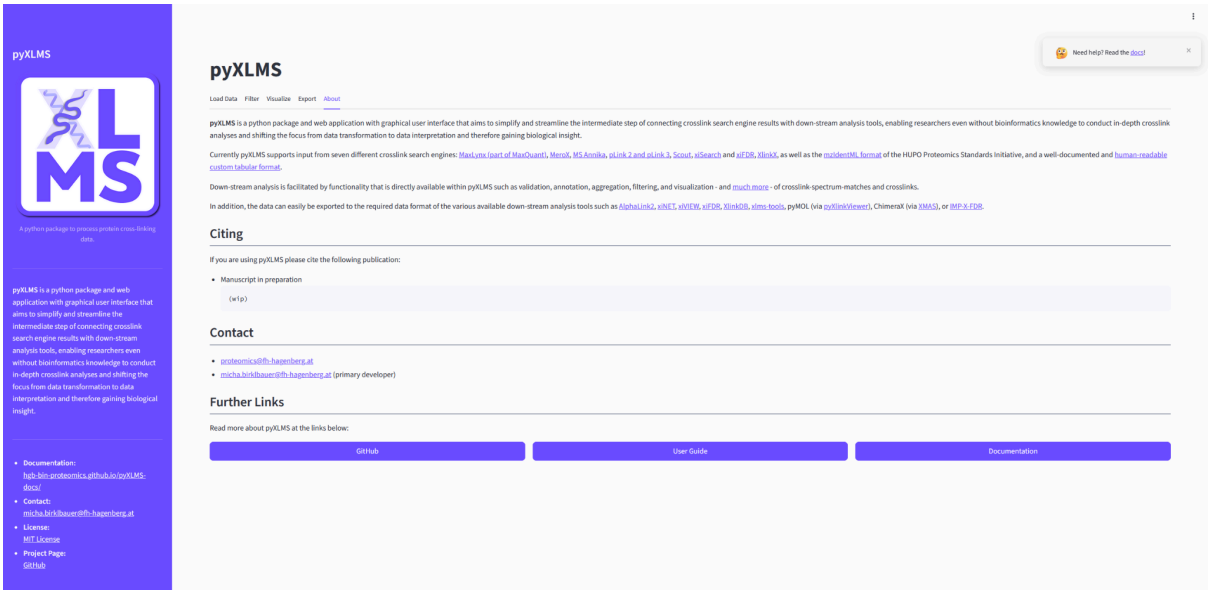
If the export finished successfully you will get the option to download the exported file(s). Sometimes also some additional information about the export is displayed.

Use Your Exported File(s) with the Selected Down-Stream Analysis Tool

For convenience the web app will also display a button that takes you directly to the selected down-stream analysis tool!

Getting Help

The About tab links some useful pages on where to learn more about pyXLMS.



TIP

If you run into any problems with the pyXLMS web application or have any questions, please check our help page on how to get support!

References

[1] Szklarczyk, D. et al. The string database in 2023: protein–protein association networks and functional enrichment analyses for any sequenced genome of interest. *Nucleic Acids Research* **51**, D638–D646 (2022).
URL http://dx.doi.org/10.1093/nar/gkac1000.

[2] Bräuer, P., Tirian, L., Müller, F., Mechtler, K. & Matzinger, M. In vivo crosslinking and effective 2D enrichment for proteome wide interactome studies. *Communications Chemistry* **8** (2025).
URL http://dx.doi.org/10.1038/s42004-025-01644-6.